

Literature Status: WCG of Free Banach Lattices over $c_0(\Gamma)$ and $\ell_p(\Gamma)$

arXiv:1706.08147 answered by arXiv:1806.02553

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Status. literature_already_answered. This packet records an exact later-paper answer, not a new proof from the run.

Original source. Antonio Aviles, Jose Rodriguez, and Pedro Tradacete, *The free Banach lattice generated by a Banach space*, arXiv:1706.08147. In Section 5, after Corollary 5.5, the paper says that it is not known whether

$$FBL[c_0(\Gamma)] \quad \text{or} \quad FBL[\ell_p(\Gamma)] \quad (2 < p < \infty)$$

are weakly compactly generated for uncountable Γ . The same paragraph notes that no non-separable Banach space E was known there with $FBL[E]$ WCG.

Supporting answer. Antonio Aviles, Pedro Tradacete, and Ignacio Villanueva, *The free Banach lattices generated by ℓ_p and c_0* , arXiv:1806.02553. The introduction explicitly says that the paper completes the $p > 2$ direction from arXiv:1706.08147. The decisive formal statement is Corollary wcg in Section 3:

For every set Γ and every $p > 2$, $FBL[\ell_p(\Gamma)]$ and $FBL[c_0(\Gamma)]$ are WCG.

Why this answers the target. The source question asks exactly for the WCG status of $FBL[c_0(\Gamma)]$ and $FBL[\ell_p(\Gamma)]$, $2 < p < \infty$, for uncountable Γ . The supporting corollary proves WCG for every set Γ , hence in particular for all uncountable Γ . The supporting authors explicitly cite the source construction and present the result as the opposite $p > 2$ counterpart to the $1 \leq p \leq 2$ non-WCG result of arXiv:1706.08147.

Scope and limitations. The packet settles only the source’s residual WCG question for the classical spaces $c_0(\Gamma)$ and $\ell_p(\Gamma)$, $2 < p < \infty$. It does not claim a characterization of all non-separable Banach spaces E for which $FBL[E]$ is WCG.

Search evidence. The local lightweight indexes contained no exact packet for arXiv:1706.08147 or arXiv:1806.02553. Web and local source search for “FBL[c_0(Gamma)] WCG”, “FBL[ell_p(Gamma)] WCG”, and “free Banach lattice weakly compactly generated c_0 ell_p” surfaced arXiv:1806.02553 as the explicit later answer.